

Reading Review: A Mean Field View of the Landscape of Two-Layer Neural Networks

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1. Motivation: Why Mean Field?

Two-layer neural networks are trained by minimizing a nonconvex high-dimensional risk function via stochastic gradient descent (SGD). The fundamental question is:

Does SGD converge to a global optimum or only to a local optimum? If the latter, why do local minima have good generalization properties?

For a two-layer network with N hidden units:

$$\hat{y}(x; \theta) = \frac{1}{N} \sum_{i=1}^N \sigma_*(x; \theta_i), \quad (1)$$

the population risk (with squared loss) admits a remarkable **mean-field decomposition**:

$$R_N(\theta) = R_{\#} + \frac{2}{N} \sum_{i=1}^N V(\theta_i) + \frac{1}{N^2} \sum_{i,j=1}^N U(\theta_i, \theta_j), \quad (2)$$

where $V(\theta) = -\mathbb{E}[y\sigma_*(x; \theta)]$ and $U(\theta_1, \theta_2) = \mathbb{E}[\sigma_*(x; \theta_1)\sigma_*(x; \theta_2)]$.

Key insight. The risk depends on parameters $\theta_1, \dots, \theta_N$ only through their **empirical distribution** $\hat{\rho}^{(N)} = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}$. This suggests studying the infinite-width limit $N \rightarrow \infty$, where the risk becomes a functional over probability measures:

$$R(\rho) = R_{\#} + 2 \int V(\theta) \rho(d\theta) + \iint U(\theta_1, \theta_2) \rho(d\theta_1) \rho(d\theta_2). \quad (3)$$

Since U is positive semidefinite, $R(\rho)$ is **convex** in ρ —a dramatic simplification!

2. Main Result: Distributional Dynamics

The central theorem of the paper establishes that SGD dynamics is captured by a nonlinear PDE in the infinite-width limit.

Theorem 1 (**Theorem 3** in paper: SGD $\rightarrow$ PDE Convergence). *Assume conditions **A1–A3** (bounded Lipschitz step size, bounded activation with sub-Gaussian gradient, bounded Lipschitz gradients of V and U). Let $\rho_0 \in \mathcal{P}(\mathbb{R}^D)$ and consider SGD with initialization $(\theta_i^0)_{i \leq N} \stackrel{iid}{\sim} \rho_0$ and step size $s_k = \varepsilon \xi(k\varepsilon)$.*

Let ρ_t solve the **Distributional Dynamics (DD)** PDE:

$$\partial_t \rho_t = 2\xi(t) \nabla_\theta \cdot (\rho_t \nabla_\theta \Psi(\theta; \rho_t)), \quad (4)$$

where $\Psi(\theta; \rho) = V(\theta) + \int U(\theta, \theta') \rho(d\theta')$.

Then for any fixed t , the empirical measure $\hat{\rho}_t^{(N)} \Rightarrow \rho_t$ almost surely as $N \rightarrow \infty$, $\varepsilon \rightarrow 0$ with $N/\log(1/\varepsilon_N) \rightarrow \infty$. Moreover, for any $f : \mathbb{R}^D \rightarrow \mathbb{R}$ with $\|f\|_\infty, \|f\|_{Lip} \leq 1$:

$$\sup_{k \in [0, T/\varepsilon]} \left| \frac{1}{N} \sum_{i=1}^N f(\theta_i^k) - \int f(\theta) \rho_{k\varepsilon}(d\theta) \right| \leq C e^{CT} \cdot \text{err}_{N,D}, \quad (5)$$

$$\sup_{k \in [0, T/\varepsilon]} |R_N(\theta^k) - R(\rho_{k\varepsilon})| \leq C e^{CT} \cdot \text{err}_{N,D}, \quad (6)$$

with probability at least $1 - e^{-z^2}$, where $\text{err}_{N,D} = \sqrt{\frac{1}{N} \vee \varepsilon(\sqrt{D} + \log(N/\varepsilon) + z)}$.

Significance.

- **Width independence:** The landscape does not become more complex as N increases. The approximation is valid when $N \gg D$ and $\varepsilon \ll 1/D$.
- **Dimensional reduction:** The $N \times D$ -dimensional SGD dynamics reduces to a D -dimensional PDE.
- **Symmetry exploitation:** The PDE formulation allows exploiting symmetries in the data distribution (e.g., rotational invariance), which is impossible for finite- N dynamics.
- **Connection to physics:** The DD PDE is a gradient flow in the Wasserstein space $(\mathcal{P}(\mathbb{R}^D), W_2)$, connecting neural network training to interacting particle systems.

3. Consequences: Noisy SGD Converges Globally

Building on Theorem 3, the paper analyzes **noisy SGD**:

$$\theta_i^{k+1} = (1 - 2\lambda s_k) \theta_i^k + 2s_k (y_k - \hat{y}_k) \nabla_{\theta_i} \sigma_*(x_k; \theta_i^k) + \sqrt{2s_k/\beta} g_i^k, \quad (7)$$

where $g_i^k \sim \mathcal{N}(0, I_D)$. The scaling limit is a **diffusion PDE**:

$$\partial_t \rho_t = 2\xi(t) \nabla_\theta \cdot (\rho_t \nabla_\theta \Psi_\lambda(\theta; \rho_t)) + 2\xi(t) \beta^{-1} \Delta_\theta \rho_t, \quad (8)$$

which is a gradient flow for the free energy $F_{\beta,\lambda}(\rho) = \frac{1}{2} R(\rho) + \frac{\lambda}{2} \int \|\theta\|^2 \rho(d\theta) - \beta^{-1} \text{Ent}(\rho)$.

Theorem 2 (Theorem 4 in paper: Convergence of Noisy PDE). *Under assumptions A1–A4 and $\lambda > 0$, the free energy $F_{\beta,\lambda}(\rho)$ has a unique minimizer $\rho_*^{\beta,\lambda}$, and the diffusion PDE converges to it: $\rho_t \Rightarrow \rho_*^{\beta,\lambda}$ as $t \rightarrow \infty$.*

Theorem 3 (Theorem 5 in paper: Noisy SGD Reaches Near-Optimum). *Under the same assumptions, for any $\eta > 0$, there exists $\beta \geq KD$ and $T < \infty$ (independent of N) such that for $k \in [T/\varepsilon, 10T/\varepsilon]$:*

$$R_N(\theta^k) \leq \inf_{\rho \in \mathcal{P}(\mathbb{R}^D)} R(\rho) + \eta \quad (9)$$

with probability at least $1 - \delta$, provided $N, 1/\varepsilon \geq C_0 e^{C_0 T} D$.

Main conclusion. Noisy SGD converges to a near-global minimum in a number of steps **independent of the network width N** .

4. Limitations and Open Questions

- **Exponential time dependence:** The error bound grows as e^{CT} , limiting applicability to cases where DD converges rapidly. The proof does not provide satisfactory estimates for the dependence of convergence time T on dimension D and tolerance η .
- **Special data distributions:** The explicit convergence results (Theorems 1–2) are for isotropic/anisotropic Gaussians—highly idealized cases.
- **Two layers only:** The analysis does not extend to deep networks with more than two layers.

1 Dependency Graph

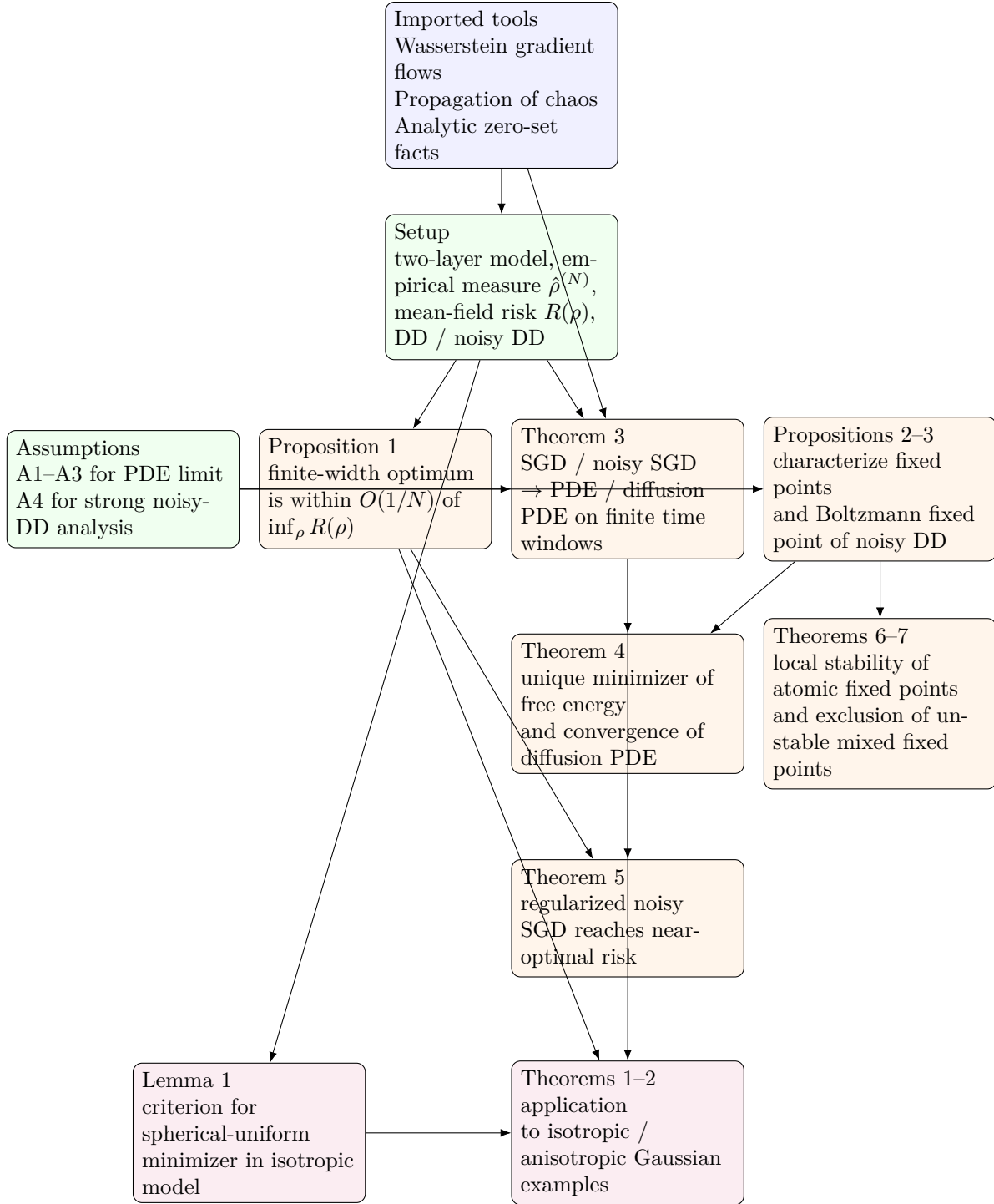


Figure 1: Main proof spine of [1]. The central transition is from finite-width SGD to distributional dynamics, after which the analysis moves to fixed points, free energy, and example-specific PDE reductions. A few logically true but visually redundant edges are omitted so the main proof spine remains readable.

How to read the graph. The paper’s most important move is Theorem 3: once SGD is approximated by DD, the optimization problem is replaced by the study of a measure-valued flow. The noisy branch (Proposition 3 $\rightarrow$ Theorem 4 $\rightarrow$ Theorem 5) gives the cleanest generic convergence result. The noiseless branch (Propositions 2, Theorems 6–7) gives a weaker but structurally informative picture of stable and unstable fixed points.

References

- [1] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.